Incremental Private Set Intersection Cardinality

Abstract

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References

[1] B. Bünz, S. Agrawal, M. Zamani, and D. Boneh. 2020. Zether: Towards Privacy in a Smart Contract World. In *International Conference on Financial Cryptography and Data Security*.

A M's zk proof

Recall in Sec. **??**, we want to prove that M knows $\mathsf{tsk}_2, \mathsf{y}, r, \{\mathsf{e_d}\}_{d=0}^{n}$ such that

$$\mathcal{B}' = \mathcal{F}(\mathsf{y}) \tag{1}$$

$$c_{\mathsf{M}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = {\mathsf{tpk}_1}^{\mathsf{tsk}_2} \wedge \tag{2}$$

$$E = \sum_{d=0}^{n}{}_{\mathsf{tpk}}\ \mathsf{e_d} \times_{\mathsf{tpk}} \vec{c}_{\mathsf{T}}[d] \wedge \tag{3}$$

$$\forall d \in [0, n],$$

$$r \neq 0 \wedge \mathsf{e_d} = r \cdot \sum_{j' \in \mathcal{B}'} {j'}^d \tag{4}$$

$$\exists\ \delta_0, \ldots, \delta_{\mathsf{n}} \in \mathbb{F}^* :$$

$$\vec{p}[d] = \vec{a}[d] \times_{\mathsf{tpk}} \mathsf{y} +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(0; \delta_{\mathsf{d}})\} \tag{5}$$

with public parameters $c_{\mathsf{M}}, E, \mathsf{tpk}, \mathsf{tpk}_1, \vec{c}_{\mathsf{T}}, \vec{a}$.

As an analogue to Bünz et al. [1], and extending their formulation to our setting with Exponential ElGamal, the statement to be proven is:

$$\{g, h, c_{\mathsf{M}}, E, \mathsf{tpk}, \mathsf{tpk}_1, \vec{c}_{\mathsf{T}}, \vec{p}, \hat{t}, \delta(y_\Sigma, z_\Sigma), \omega_{\mathsf{y}}, \omega_r, \{\omega_d\}_{d=0}^{n};$$
$$\mathsf{tsk}_2, \mathsf{y}, \{\mathsf{e_d}\}_{d=0}^{n}, r, \{\delta_{\mathsf{d}}\}_{d=0}^{n}, \tau |$$
$$c_{\mathsf{M}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = {\mathsf{tpk}_1}^{\mathsf{tsk}_2} \wedge$$
$$E[1] = g^r \prod_{d=0}^{n} \vec{c}_{\mathsf{T}}[d][1]^{\mathsf{e_d}} \wedge E[2] = \mathsf{tpk}^r \prod_{d=0}^{n} \vec{c}_{\mathsf{T}}[d][2]^{\mathsf{e_d}} \wedge$$
$$\forall d \in [0, n], \vec{p}[d][1] = (g)^{\delta_{\mathsf{d}}} (\vec{a}[d][1])^{\mathsf{y}} \wedge$$
$$\vec{p}[d][2] = (\mathsf{tpk})^{\delta_{\mathsf{d}}} (\vec{a}[d][2])^{\mathsf{y}}$$
$$\wedge\ r \neq 0 \wedge \forall d \in [0, n], \mathsf{e_d} = r \cdot \sum_{j' \in \mathcal{B}'} {j'}^d$$
$$\wedge\ t = \hat{t} - \delta(y_\Sigma, z_\Sigma) \wedge g^{t - \omega_{\mathsf{y}} r - \omega_r r - \sum_{d=0}^{n} \omega_d \mathsf{e_d}} h^\tau = T_{1,2}\}$$

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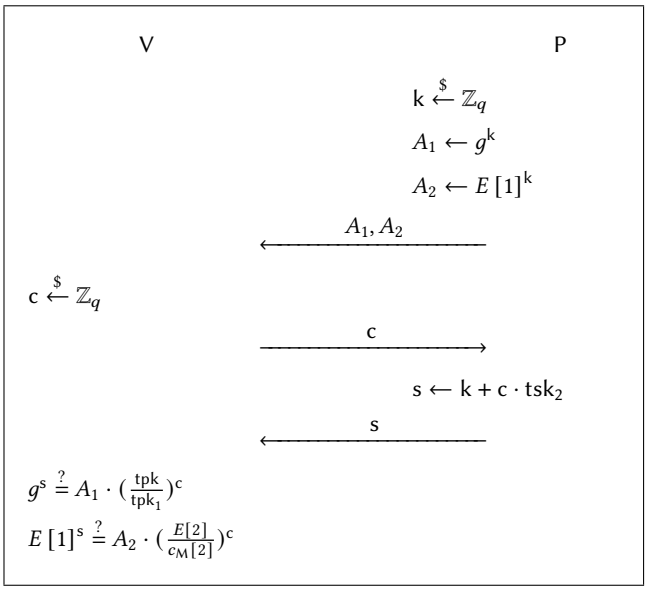


Figure 1: Proof π_{dec} for (2): $c_{\mathsf{M}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = {\mathsf{tpk}_1}^{\mathsf{tsk}_2}$

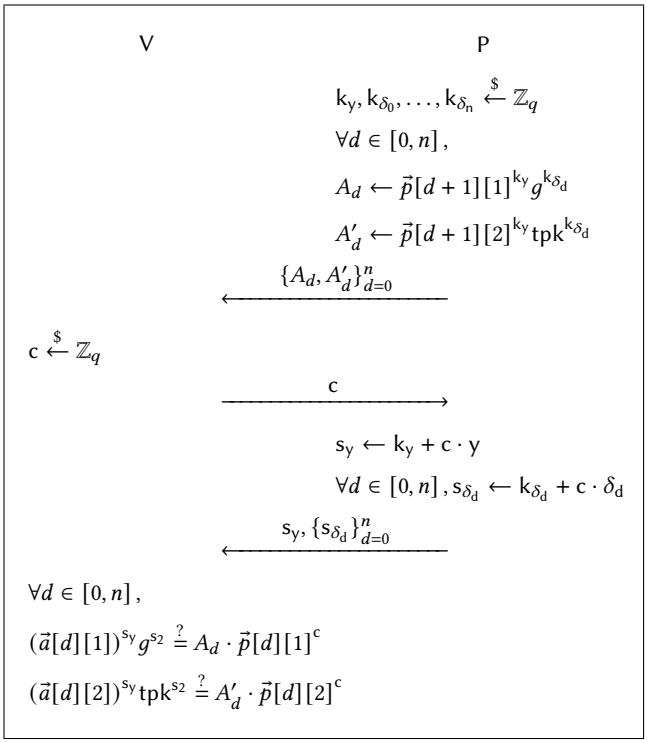


Figure 2: Proof π_{next} for (5): $\vec{p}[d] = \vec{a}[d] \times_{\mathsf{tpk}} \mathsf{y} +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(0; \delta_\mathsf{d})$

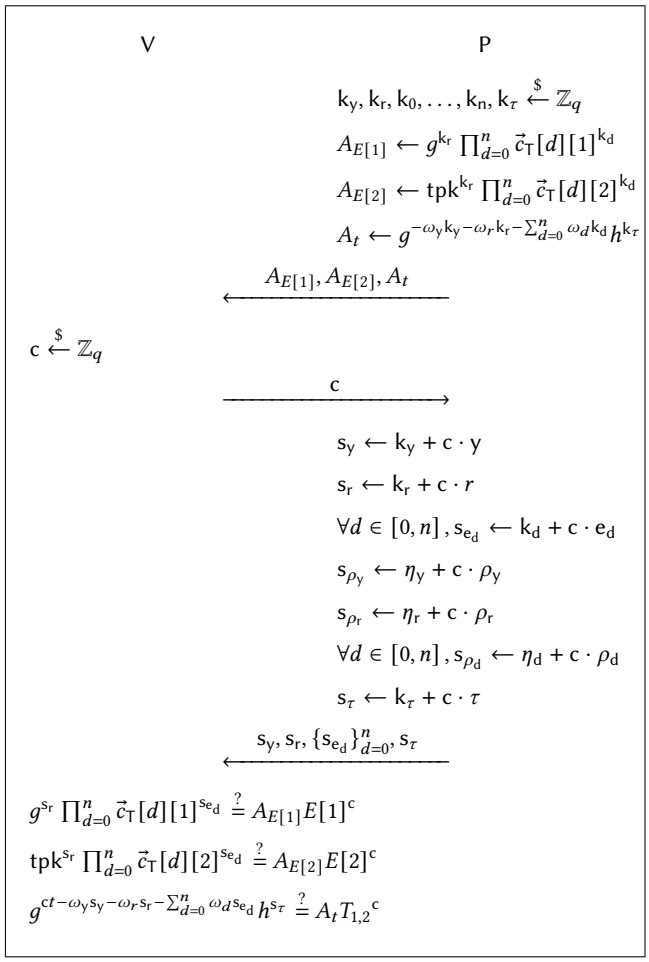


Figure 3: Proof π_Σ for Σ-Bullets Conjunction of (3) and (1), (4): $E = \sum_{d=0}^{n}{}_{\mathsf{tpk}}\ \mathsf{e_d} \times_{\mathsf{tpk}} \vec{c}_{\mathsf{T}}[d] \wedge \forall d \in [0,n]\,, \mathsf{e_d} = r \cdot \sum_{j' \in \mathcal{B}'} j'^d \wedge r \neq 0$